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-- Engineer: Olasubomi Borishade  
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-- Create Date: 05/29/2026 11:23:00 PM  
-- Design Name: MAC Unit  
-- Module Name: MAC_Unit - Behavioral  
-- Project Name: Parameterized Matrix Multiply Engine  
  
-- Description: Fundamental digital circuit that computes the  
product of  
--                two numbers and adds the result to an internal  
accumulator.  
--  
-- Revision:  
-- Revision 0.01 - File Created  
-- Additional Comments:  
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library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;  
use IEEE.NUMERIC_STD.ALL;  
  
-- Uncomment the following library declaration if instantiating  
-- any Xilinx leaf cells in this code.  
--library UNISIM;  
--use UNISIM.VComponents.all;  
  
entity MAC_Unit is  
  
Generic (    Input_Width          : INTEGER := 8;  
            Accumulator_Width    : INTEGER := 32  
            );
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Port      (      Input_A          : in SIGNED (Input_Width - 1 downto
0);

          Input_B          : in SIGNED (Input_Width - 1 downto
0);

          Clock            : in STD_LOGIC;
          Reset            : in STD_LOGIC;
          Enable           : in STD_LOGIC;
          Clear_Accumulator : in STD_LOGIC;
          Output           : out SIGNED (Accumulator_Width - 1
downto 0)
          );

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end MAC_Unit;

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architecture Behavioral of MAC_Unit is

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Signal Product          : SIGNED ((2 * Input_Width) - 1
downto 0);
Signal Accumulator_Register : SIGNED (Accumulator_Width - 1
downto 0);

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begin

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Product <= Input_A * Input_B;

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Process(Clock)

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begin

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    if rising_edge(Clock) then
        if Reset = '1' then
            Accumulator_Register <= (others => '0');
        elsif Clear_Accumulator = '1' then
            Accumulator_Register <= (others => '0');
        elsif Enable = '1' then
            Accumulator_Register <= Accumulator_Register +
resize(Product, Accumulator_Width) ;
        end if;
    end if;

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end process;

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Output <= Accumulator_Register;
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end Behavioral;
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